TrailMap

Offline Topographic Map & Track Recorder

User Manual — Version 1.9

TrailMap is an Android app for offline topographic mapping, GPS track recording, and 3D terrain visualization. It uses Mapsforge vector maps with OpenAndroMaps data and SRTM elevation data for accurate 3D terrain rendering. Designed for hiking, mountain biking, and motorcycling.

Contents

1. Installation

1.1 Requirements

- Android 8.0 (Oreo) or later
- At least 2 GB free storage (more recommended for maps)
- GPS-capable Android device
- Internet connection required for initial map and elevation downloads only

1.2 Installing the APK

TrailMap is distributed as an APK file. Since it is not on the Google Play Store, you must enable installation from unknown sources.

Step 1 — Enable Unknown Sources

- Open **Settings** on your Android device
- Go to **Security** (or **Privacy** on some devices)
- Enable **Install unknown apps** or **Unknown sources**
- On Android 8+: go to **Settings > Apps > Special app access > Install unknown apps** and enable it for your file manager or browser

Step 2 — Transfer the APK

- Copy the **TrailMap.apk** file to your device via USB, email, Google Drive, or direct download
- The easiest method is to connect via USB and copy to the Downloads folder

Step 3 — Install

- Open your file manager and navigate to the APK file
- Tap the APK file
- Tap **Install** when prompted
- Tap **Open** to launch TrailMap

Step 4 — Grant Permissions

On first launch, TrailMap will request the following permissions:

- **Location (Fine)** — required for GPS tracking
- **Storage / Files** — required for saving maps, tracks, and elevation data

> *Note: If you deny a permission, the app will ask again next launch. Permissions can also be granted later via Settings > Apps > TrailMap > Permissions.*

1.3 Updating the App

To update TrailMap, simply install a newer APK over the existing installation. Your maps, tracks, waypoints, and elevation data are preserved during updates.

2. Getting Started — Downloading Maps

TrailMap works entirely offline once maps are downloaded. Maps must be downloaded before use. An internet connection is required only for downloading.

2.1 Opening Map Manager

- Tap the **menu icon** (three dots) in the top-right corner of the main screen
- Select **Map Manager**

2.2 Downloading a Map

- The Map Manager lists US states and Canadian provinces
- Tap the **download icon** next to your region
- Three files are downloaded in sequence:

  Map — the vector map file (.map)

  POI — points of interest for search (.sqlite)

  Elev — SRTM elevation tiles (.hgt.zip)
- Progress shows as **Map: downloading...**, **POI: downloading...**, **Elev: N/M** (N tiles of M complete)
- Tap a downloaded map to **activate** it (shown with checkmark)

> *Tip: Download elevation data for accurate 3D terrain. Without elevation data, the 3D viewer uses GPS elevation which is less accurate.*

2.3 Deleting a Map

- Tap the **delete icon** next to a downloaded map
- Elevation tiles shared with other downloaded maps are preserved automatically

2.4 Purple 'Elev' Button

If a map is downloaded but elevation data is missing, a purple **Elev** button appears next to it in Map Manager. Tap it to download elevation data separately.

3. Map Navigation

3.1 Basic Controls

Gesture	Action
One finger drag	Pan the map
Two finger pinch/spread	Zoom in / zoom out
Two finger rotate	Rotate map
Double tap	Zoom in
Long press	Add a waypoint at that location

3.2 Toolbar Buttons

Button	Function
Location arrow	Center map on current GPS position
Record (circle)	Start GPS track recording
Stop (square)	Stop recording and save track
Tracks (list icon)	Open track list
Search (magnifier)	Search for POIs
Waypoints (flag)	Open waypoint list
Satellites (signal)	Show satellite status
Menu (three dots)	Access Map Manager and settings

3.3 GPS Indicator

The GPS status is shown in the top bar. A green dot indicates a good fix. The satellite display shows the count and average signal strength, e.g. **31 sats / 24 dB**.

4. Recording a Track

4.1 Starting a Recording

- Ensure GPS has a fix (green indicator)
- Tap the **Record** button (circle icon) in the toolbar
- The button turns red and a recording indicator appears
- TrailMap records your position, elevation, and timestamp continuously

> *Tip: Keep the screen on or plug in to a charger to prevent the recording from stopping. TrailMap auto-enables screen-on when charging.*

4.2 During Recording

- An orange arrow shows your current position and heading on the map
- The map follows your position automatically
- Elapsed time and distance are shown in the status bar

4.3 Stopping and Saving

- Tap the **Stop** button (square icon)
- The track is automatically saved as a GPX file
- If SRTM elevation data is available, accurate elevation from the terrain map is saved to the GPX file — replacing GPS elevation which can have 30–50 ft of error
- Track files are stored in the **Tracks** folder on internal storage

4.4 Track File Format

Tracks are saved as standard GPX 1.1 files and are compatible with Garmin, Strava, Komoot, RideWithGPS, and other applications. Elevation values in the GPX are in meters.

5. Track Management

5.1 Opening the Track List

- Tap the **Tracks** icon in the toolbar
- Tracks are listed with date, time, distance, and elevation gain
- Tap a track to open its options

5.2 Track Options

Tapping a track shows three options:

- **View 3D** — Opens the 3D terrain viewer with the track overlaid on the map and elevation data
- **Rename** — Rename the track with a custom name
- **Delete** — Permanently delete the track (with confirmation)

5.3 Viewing a Track in 3D

When View 3D is selected:

- The map automatically zooms and pans to the track area
- If the track is wider than tall, the phone rotates to landscape for a better capture
- The map tiles fully load before the capture is taken
- The 3D viewer opens with the track rendered on SRTM terrain

> *Note: The map captures the area at the current zoom level. The phone returns to portrait orientation after the viewer launches.*

6. 3D Track Viewer

The 3D Track Viewer uses WebGL to render a true 3D terrain model with the map texture draped over SRTM elevation data and the GPS track overlaid on the surface. All rendering happens on the device GPU for smooth performance.

6.1 Initial View

The viewer opens in a birds-eye view looking slightly down from above. The track is oriented to best fill the screen — wide tracks open rotated 90 degrees. The **Reset** button (circular arrow) returns to this default view.

6.2 Controls

Control	Action
One finger drag	Rotate the terrain
Pan button (active)	One finger drag pans instead of rotates
+ button	Zoom in
- button	Zoom out
Pan/Rotate toggle	Switch between pan and rotate mode
Reset button (arrow)	Return to default view angle
2D button	Switch to 2D elevation profile
Elevation Scale slider	Exaggerate or flatten terrain height

6.3 Track Colors

The track is color-coded by elevation:

- **Blue** — lowest elevation on the track
- **Magenta/Pink** — mid elevation
- **Red** — highest elevation on the track

The elevation bar on the right side of the screen shows the color scale with exact elevation values at top (max) and bottom (min).

6.4 Terrain Rendering

- The terrain mesh is a 24x24 grid built from SRTM 30-meter elevation data
- The map texture is draped over the terrain
- Brown skirts on all four edges show the terrain cross-section
- The track sits approximately 25ft above the terrain surface to remain visible

6.5 Elevation Scale

The **Elev Scale** slider on the right controls vertical exaggeration:

- **1x** — true scale (terrain may appear flat for gentle terrain)
- **2x** — default, good for most trails
- **5x** — maximum, useful for visualizing subtle elevation changes

6.6 Compass

A compass rose appears next to the stats box. The red arrow always points north and rotates as you spin the terrain view. The **N** label moves with the arrow tip.

6.7 Elevation Accuracy

TrailMap uses SRTM elevation data for both the terrain mesh and the track elevation, so they are from the same source and match accurately. GPS elevation (which can be 30–50 ft off) is replaced by SRTM elevation when a track is saved.

7. 2D Elevation Profile

Tap the **2D** button in the 3D viewer to switch to a 2D elevation profile chart. Tap **3D** to return.

- The chart shows elevation (ft) on the Y axis and distance (miles) on the X axis
- The track is color-coded by elevation using the same blue→magenta→red scale
- A green dot marks the start, red dot marks the end
- Touch or drag across the chart to show a crosshair with distance and elevation readout

> *Tip: The 2D profile shows the SRTM-corrected elevation, so the start and end of a loop ride will show the same elevation even if GPS drift would have shown a difference.*

8. Waypoints

8.1 Adding a Waypoint

- Long-press anywhere on the map
- An info window appears with the coordinates
- Tap **Add Waypoint** in the info window
- Enter a name and tap OK

8.2 Navigating to a Waypoint

- Tap the **Waypoints** flag icon in the toolbar
- Tap **Go** next to a waypoint
- A navigation panel appears at the bottom showing:
  Bearing — direction to waypoint in degrees
  Distance — straight-line distance to waypoint
- Tap the navigation panel to stop navigation
- The map snaps back to following your GPS position when navigation ends

8.3 Deleting a Waypoint

- Open the Waypoints list
- Tap **Delete** next to a waypoint
- The waypoint marker is removed from the map immediately

9. POI Search

TrailMap can search for points of interest within the downloaded map area. Search works fully offline using the POI database downloaded with each map.

- Tap the **Search** magnifier icon in the toolbar
- Type a name or category (e.g. *campground*, *gas station*, *hospital*)
- Results appear as a list — tap a result to center the map on it
- A marker is placed at the POI location

> *Note: Search results are limited to the currently active map region. Switch maps in Map Manager if searching a different area.*

10. Satellite Status

Tap the **satellite/signal icon** in the toolbar to view GPS satellite status.

- Shows total satellites in view and average signal strength in dB
- Example: **31 sats / 24 dB**
- Higher dB = stronger signal = better position accuracy
- Typical good fix: 8+ satellites, 20+ dB

> *Tip: Wait for a good satellite fix before starting a track recording to ensure accurate positioning from the start.*

11. Settings & Tips

11.1 Screen-On While Charging

TrailMap automatically keeps the screen on when the device is charging. This is useful when the phone is mounted on handlebars with a charger. The screen will go to sleep normally on battery to conserve power.

11.2 Map Rotation

The map can be freely rotated with two fingers. The map always renders with north-up for map capture before the 3D viewer launches, regardless of current rotation.

11.3 Storage Locations

Data Type	Location
Map files (.map)	Internal storage / TrailMap / maps /
POI databases (.sqlite)	Internal storage / TrailMap / maps /
Elevation tiles (.hgt)	Internal storage / TrailMap / elevation /
Track files (.gpx)	Internal storage / TrailMap / tracks /

11.4 Sharing Tracks

GPX files can be copied from the tracks folder and shared with any app or service that supports GPX format. Connect via USB and browse to **Internal storage / TrailMap / tracks /**.

11.5 Battery Tips

- Use a handlebar mount with USB charger for long rides
- Screen-on mode activates automatically when charging
- GPS recording continues in the background if the screen turns off
- Reduce screen brightness to extend battery life

11.6 Troubleshooting

Problem	Solution
Map not showing	Open Map Manager and activate a downloaded map
No GPS fix	Go outdoors with clear sky view and wait 30–60 seconds
3D viewer blank	Ensure elevation data is downloaded for the map region
Track missing elevation	Re-record track — new tracks use SRTM elevation automatically

Track buried in terrain	Track uses SRTM elevation + 25ft clearance; if still buried, the terrain mesh may have a h
Map tiles not loading	Download maps via Map Manager; internet required for download only